

Exact time is not important. we care about the growth rate ~~with~~ not exact value

Tips remember

Asymptotic notation ignores constants and lower-order terms and features on the dominant term

Dominant $\rightarrow$ It is part of complexity function that grows the fastest as input size n becomes very large

It is way to describe the characteristics of a function in the limit.

- $\rightarrow$ It describes the rate of growth of function.
- $\rightarrow$ It is a way to compare size of functions.

There are five asymptotic notations.

Big O $H(O)$ $\rightarrow$ It is generally used to represent the worst case

Big Ω $MEGA (\Omega)$ -

Big Θ notation tells us the maximum (upper bound) growth of an algorithm.

Big Θ notation the

Big Θ = highest power of n

It is generally used to define the worst case condition and define maximum upper bound